Reg. No									

St. Joseph's College of Arts & Science (Autonomous)
St. Joseph's College Road, Cuddalore – 607001

PPH807S – QUANTUM MECHANICS - I

Time : 3 hrs **Max Marks :75**

SECTION – A (10X2=20)

Answer **ALL** Questions

1. State Ehrenfest's theorem.
2. Define eigen value.
3. What is a rectangular barrier?
4. Define rigid rotator.
5. Provide the commutation rules of angular momentum operator.
6. Spin matrix-define.
7. What do you mean by perturbation?
8. Explain the WKB approximation?
9. Give the relation between probability density and current density.
10. What are gamma matrices?

SECTION – B (5X5=25)

Answer **ALL** Questions

11. a) What are Hermitian operators? Prove that the eigenvalues of a Hermitian operator are real.

(or)

b) Explain orthogonal and orthonormal functions.

12. a) Discuss the theory of finite square well.

(or)

b) Obtain Schrodinger equation for a linear harmonic oscillator.

13. a) Write a note on ladder operators.

(or)

b) Show that the eigen values of $L_z = m\hbar$

14. a) Discuss the variational method of approximation.

(or)

b) Explain Stark Effect in Hydrogen atom.

15. a) Obtain the covariant form of Dirac's equation.

(or)

b) Describe the Dirac concept of positive and negative energy states of an electron.

SECTION –C (3X10=30)

Answer any **THREE** Questions

16. Explain any two postulates of quantum mechanics.

17. Solve the Schrodinger equation for the case of hydrogen atom.

18. Discuss the angular momenta and Clebsch Gordon coefficients.

19. Using Non degenerate perturbation, obtain the ground state energies of a anharmonic oscillator.

20. Derive Klein-Gordan equation.

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